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Multi-stage knowledge distillation with layer fusion-based deep learning approach for skin cancer classification

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Skin cancer is one of the most common types of cancer globally, caused by prolonged exposure to the sun's UV rays. Despite recent developments in research, early diagnosis, prevention, and treatment, skin cancer remains a significant health concern. This study proposes a multi-stage knowledge distillation-based deep learning technique with a layer fusion strategy to classify different types of skin lesion cells using the HAM10000 dataset. Augmentation and basic preprocessing steps have been applied to the HAM10000 dataset to enhance robustness. The applied multi-stage knowledge distillation incorporates intermediate features by measuring several loss values and coefficients to balance the corresponding losses. The proposed ViT and ConvNeXT-integrated teacher model leverages hybrid architectures derived from baseline models, combining convolutional feature extraction with transformer-based attention mechanisms. The distilled model, built using CNN and EfficientNet, achieved significant performance improvements over the baseline. The optimized model achieved an accuracy of 95.88%, F1 and AUC scores of 95.91% and 99.02%, respectively. Multi-stage knowledge distillation with intermediate exits and layer fusion improved model accuracy and performance metrics with the lowest training and inference times of 61 and 15 ms/step, respectively. Post-training quantization was applied to reduce the parameters and size of the distilled model. Multiple XAI techniques—Grad-CAM, Score-CAM, and LIME—have been explored to enhance the interpretability of the applied multi-stage knowledge distillation model. The implementation codes with the quantized and distilled model are available in the following repository: https://github.com/codewith-pavel/Optimizations.

Skin cancer is a severe disease that occurs from the uncontrolled growth of several cells in the body[1]. It is the most common cancer in the world, mainly affecting men. This oncological disease impacts 1 in 5 people and more than two people die every hour[2]. Most cases of skin cancer are due to ultraviolet rays from the sun. These UV rays can damage skin cells and cause sunburn[3]. The UV rays from the sun, sunlight, or tanning beds come across the skin cells to grow more quickly. Several risk factors are dangerous for enabling skin cancer, including light skin, age, tobacco smoking, consumption of alcohol, HPV infections, and others[4].

The most common symptom of skin cancer is visible changes in the skin cells[5]. Another symptom includes a mole in the skin that changes in size, shape, and color or that bleeds. Skin cancers occur commonly in women before the early age of 50. However, it appears frequently in men of more than 50 ages. Skin cancer is often diagnosed in later stages, which becomes difficult to treat[6]. A sample of the malignant tissues is removed and sent to the laboratory for testing under a microscope. The skin cancer treatment depends on the stage of the cancer[7]. Liquid nitrogen is enabled in Cryotherapy to freeze the tissue. After treatment, the dead cells slough off. The excisional surgery procedure is about removing the tumor. Elimination is performed only on affected cells during Mohs surgery. Medications are valuable and used to kill cancer cells in chemotherapy. A dermatologist performs all these treatment techniques. The radiation oncologist uses radiation to kill cells or force them to avoid the unconditional growth of cells.

Early skin cancer detection can help protect the body from diseases and cure it immediately. Modern methods, including machine learning and deep learning in the AI field, have progressed much in classifying and detecting diseases earlier than traditional methods. Artificial Intelligence has shown great promise in healthcare subjects,

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especially in diagnosing skin cancer[8]. The modern method will reduce the complexity of previous methods and can improve the skin cancer diagnosis rate. The existing literature suggests that AI systems can classify, detect, and segment the different types of skin cancer better than dermatologists. This work represents skin cancer detection with advanced AI techniques. In this study, we propose a lightweight and accurate skin cancer classification framework using a multi-stage knowledge distillation approach. The method transfers knowledge from a hybrid transformer-based teacher model (ViT + ConvNeXT) to a hybrid CNN-based student model (CNN + EfficientNet). We introduce intermediate exits and layer fusion during training to further enhance performance and generalization. The final model is optimized with post-training quantization and supported by explainable AI techniques for interpretability. The contributions of this work are as follows:

- An innovation of the multi-stage knowledge distillation-based deep learning approach that distills knowledge between multiple output layers of the teacher and student model and takes advantage of learning inside hidden fused layers.
- An ensemble approach with ViT and ConvNeXT Transformer to make a hybrid transformer teacher model for observing the comparison with a hybrid student model developed with a combination of CNN and EfficientNet techniques and layer fusion strategy to aggregate outputs for better prediction and performance.
- The post-training quantization reduces the model's weight and activations from high to low-precision formats to represent the applied multi-stage distilled model's parameters.
- The ISIC 2018 test data set has been used to assess the generalization and clinical relevance of the proposed MSKD framework, including the hybrid student baseline (HSB), knowledge distillation model (KD) and post-training quantized model (PTQ).
- The implementation of multiple XAI techniques, including Score-CAM, Grad-CAM, and LIME, as explainable AI to visualize identification and validation of the results.

The novelty of this study lies in integrating optimization, quantization, and knowledge distillation with modified layer fusion in a hybrid model while incorporating XAI techniques for enhanced interpretability.

The remaining sections of the article are structured as follows: Section describes the summary from prior studies, mentioning their limitations and introducing the innovative approach adopted in this study. Section 2 explains the system architecture and implementation details. Section 3 presents the outcomes of applying baseline models, multi-stage knowledge distillation, and explainable AI visualization. Section 4 concludes this work's overall performance and showcases changeovers after several optimization and layer fusion techniques.

Background studies

In recent studies, numerous research has progressed to advance the predictive capabilities for advanced medical image analysis, including identifying various types of brain tumor[9–11], breast cancer[12], skin cancer[13], gastrointestinal disease[14,15], brain stroke[16], etc. The symptoms, treatment, and traditional diagnosis of skin cancer face challenges concerning subjective assessment due to manual inspection, costs, time consumption, and resource availability[17]. The primary focus of these studies includes the classification, segmentation, and detection of skin cancer types with traditional CNN and modern vision transformer architectures. A brief overview of recent articles is presented in the subsequent paragraphs.

Traditional CNN architecture

Farea et al.[18] developed a hybrid AI framework for skin cancer detection with combined diverse public datasets. The authors applied multiple CNN models, i.e., DenseNet121, InceptionResNetV2, etc., for further comparison with their proposed model. The proposed framework attained 93.04% accuracy and 93.12% F1-score. Faghihi et al.[19] performed skin lesion classification with CNN models. They developed a novel architecture combining VGG16 and VGG19 frameworks into a modified AlexNet network with subject-specific datasets with 2541 images. Various optimization techniques, including dropout and k-fold cross-validation, are integrated with the hybrid deep learning model to improve performance. The introduced model improved accuracy by approximately 3% (94% to 97.18%). Saleh et al.[20] developed various models and aimed to select the optimum model via the RAPS method and the ISIC 2017 dataset. The authors employed CNNs for feature extraction with a Grey Wolf optimizer and six classifiers for classification purposes. The study obtained 94.5% accuracy with the RAPS-integrated optimum model. Gamil et al.[21] used the principal component analysis technique for dimension reduction and different models, AdaBoost and EfficientNet B0, for better performance algorithms. The developed model achieved an accuracy of 93% on the DermIS dataset and 91% on the ISIC dataset.

Deepa et al.[22] performed skin cancer prediction with the modified CNN. CNN was integrated with hierarchical features to capture patterns. The modified CNN attained 98.25% accuracy, 97.32% precision, 97.10% recall, and 96.92% F1-score. Kandhro et al.[23] applied VGG19 (E-VGG19), an efficient skin cancer classifier model combining transfer learning (pre-trained) and machine learning classifiers. The E-VGG19 model achieved high classification performance, with a precision of 0.87 for benign and 0.94 for malignant cases, recall values of 0.93 and 0.86, and F1-scores of 0.89 and 0.87, respectively. Akinrinade and Du[24] implemented deep learning techniques (i.e., Convolutional Neural Networks) for early skin cancer detection. The authors applied generative adversarial networks to address the imbalanced dataset problem. Their study investigates different types of deep learning models and shows that ensemble and hybrid methods improve classification performance in the differentiation between benign and malignant skin lesions. Asif et al.[25] used the HAM10000 dataset with augmentation to classify seven skin lesion types. The authors introduced a lightweight deep learning model, SKINC-NET, which outperformed ResNet50, VGG16, MobileNetV2, and EfficientNetB0, achieving an accuracy of 98.54%.

Modern transformer architecture

Ji et al.[26] introduced EFAM-Net, a novel classification model. EFAM-Net contains three novel architectures named ConvNeXt (ARLC) block, a Parallel ConvNeXt (PCNXt) block, and a Multi-scale Efficient Attention Feature Fusion (MEAFF) block. These three blocks extract complex feature information from skin lesion images. The applied EFAM-Net obtained 92.30%, 93.95%, and 94.31% accuracies, respectively, on the HAM10000 public dataset. Reis et al.[27] built DSCIMABNet with multi-head attention depthwise separable convolution techniques, utilizing an ensemble deep learning method. The DSCIMABNet hybrid method accomplished 99.40% accuracy on the ISIC 2018 dataset. Natha et al.[28] applied the maximum voting of various pre-trained deep learning networks to optimize skin cancer classification. The applied model attained 94.70% accuracy.

Khan[29] operated two datasets, the MHC-CT dataset, including 1860 tumor and 1024 normal CT scans for binary classification, and a publicly available CT scan dataset for multiclass classification. He presented a deep learning framework with feature fusion, Inception blocks, and ConvLSTM and evaluated RestNet50 and VGG19 for feature extraction, acquiring 99.60% and 91.31% accuracy, respectively. Bilal et al.[30] used the Mendeley LBC (4-class) and BloodMNIST (8-class) datasets, employing DenseNet169, DenseNet201, and Xception with CBAM for feature enhancement and SSA for ensemble weight optimization, achieving 99.48% and 95.23% accuracy, respectively. Khan et al.[31] utilized public multi-class medical image datasets, including brain MRI, chest X-ray, and gastro-endoscopic images, to evaluate a truncated lightweight MobileNet-based model with naive inception block, ConvLSTM, and metaheuristic optimization (Chain Foraging and Cyclone Aging). The model accomplished an overall accuracy of 97.37%. Khan et al.[32] used two publicly available datasets, BR35H (binary classification, 3000 images) and Figshare (multiclass classification, 3063 images). The authors applied a modified MobileNet model enhanced with a residual skip block, separable convolution layers, and Swish activation, achieving an accuracy of 99.83% and 96.95%, respectively.

Karthik et al.[33] developed a hybrid classification system assembling two tracks, i.e., Swin transformer and dense group shuffle non-local attention (DGSNLA) network. A Swin transformer was used in the first track to extract long-range dependencies, and the DGSNLA network combined DenseNet-169, group shuffle depth-wise blocks, and enhanced non-local attention blocks. The proposed network achieves an accuracy of 94.21% and an F1-score of 96.64% on the HAM10000 dataset. Di et al.[34] investigated the ECRNet to enhance the capability of skin cancer image recognition. Experimental results show that the ECRNet model performs better than the CNN and Vision Transformer methods. The model showed 1.19% and 3.28% improvements in accuracy and F1-score, respectively, in the ISIC 2018 dataset.

Various machine learning and deep learning frameworks have been employed to classify different types of skin cancer. However, these studies exhibit several limitations, including the absence of hybrid models, imbalanced datasets, a high number of trainable parameters, a lack of optimization techniques, the omission of explainable AI methods, and limited validation by domain experts. This study addresses these challenges by introducing an optimized multi-stage knowledge distillation approach and selecting hybrid models to enhance the prediction of skin cancer types. Additionally, it incorporates various explainable AI strategies to validate the outcomes of the distilled models.

Experimental design and methods

Advancements in the AI field have introduced new possibilities for skin cancer diagnosis by enabling quicker and more accurate results. Fig. 1 outlines the comprehensive process for the proposed multi-stage knowledge distillation-based skin cancer diagnosis. The HAM10000 dataset served as the foundation for skin lesion analysis. Exploratory data analysis implementation is initiated in the next step for statistical analysis to visualize the dataset and analyze class distributions for various purposes, i.e., outliers detection, checking dataset quality, and providing crucial insights for performing image preprocessing. The dataset preprocessing includes resizing, normalizing, formatting, and other steps. Multiple deep learning models, such as Convolution-based and Transformer-based architectures, were explored in the study. The selected models have merged into a hybrid teacher-student framework for multi-stage knowledge distillation. The outcome of the fine-tuned knowledge distillation methods was the distilled student model by tuning the corresponding hyperparameters. Post-training quantization was applied to reduce the size of the distilled model. Explainable AI methods, including Grad-CAM, Score-CAM, and LIME, provide explainability by highlighting the detected regions for skin lesion prediction. The distilled model is further optimized to aid clinicians in diagnosing and managing skin lesions.

Dataset preprocessing

This work uses the HAM10000 (Human Against Machine with 10000 training images) dataset[35] comprising 10015 images of dermatoscopic types. The study also utilizes the ISIC 2018 test dataset[36], a well-established benchmark in skin lesion analysis, to evaluate the generalization performance and clinical applicability of the proposed Multi-Stage Knowledge Distillation (MSKD) framework under real-world dermatological imaging scenarios. Fig. 2 presents the distribution of different skin cancer types in the HAM10000 dataset, revealing a significant class imbalance among the categories. Melanocytic nevi constitute the majority class with 6705 instances, while other cancer types, such as dermatofibroma (115) and vascular lesions (142), are notably underrepresented. Melanoma and Benign keratosis-like lesions categories consist of 1113 and 1099 images, respectively. The images were resized to 64×64 dimensions and normalized by dividing by 256. The skin lesion dataset contained a few null values in the age column, which were handled by replacing them with the column's mean value. The lesion types were renamed using a predefined dictionary mapping.

Exploratory data analysis

Exploratory Data Analysis (EDA) examines the characteristics, patterns, and relationships within a dataset. It was performed on the HAM10000 dataset to understand its structure and identify relationships based on factors

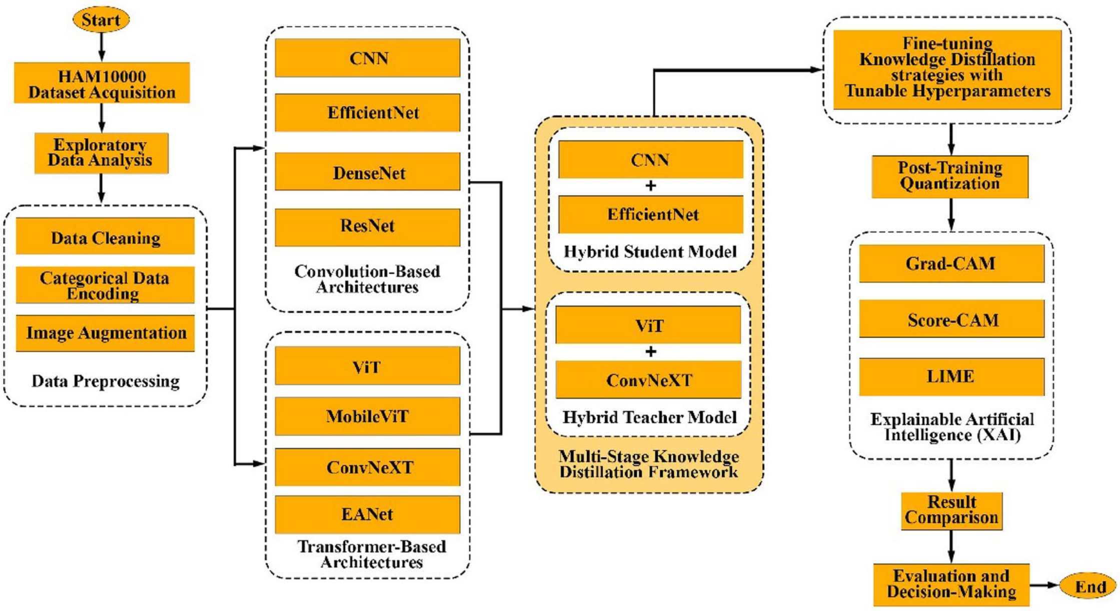


Fig. 1. The overall block diagram of the entire study.

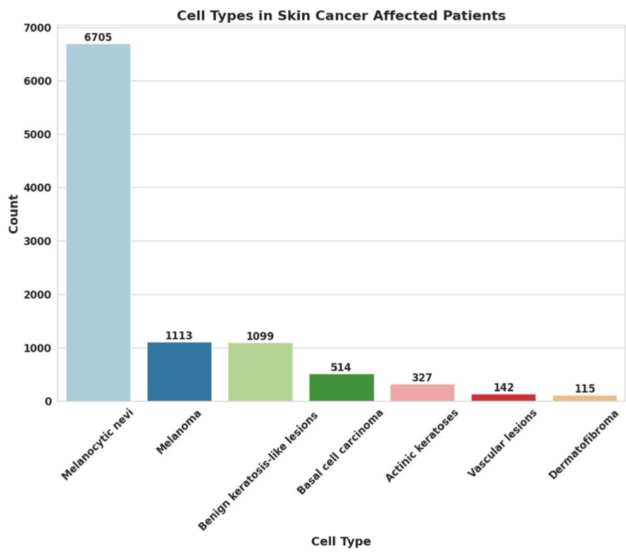


Fig. 2. Frequency of different skin cancer types of the Employed HAM10000 dataset.

such as age and gender. EDA also provides valuable insights and techniques for improving data quality, guiding model selection and enhancing performance robustness.

1. **Age**: Skin cancer is most commonly observed in individuals over the age of 50. Fig. 3 presents a histogram illustrating the age distribution, which appears bell-shaped in middle age and skewed toward older individuals. The highest frequency is observed between the ages of 40 and 50, with over 1,200 recorded observations. This distribution is particularly relevant for studying age-related patterns in skin lesions. The HAM10000 dataset provides a well-balanced representation across age groups, with missing values in the age column imputed using the mean. The age range in the dataset spans from near 0 to below 90 years, with fewer individuals observed in the younger (0–20 years) and older (above 70 years) age groups.
2. **Gender**: Skin cancer occurs more frequently in males than in females, which may be attributed to increased outdoor exposure to UV radiation. Fig. 4apresents a bar chart depicting the gender-wise distribution in the HAM10000 dataset. The dataset includes 5,406 male and 4,552 female individuals, with a small group of 57 individuals classified under the unknown category. Fig. 4bpresents a bar chart depicting the distribution of distinct cell types across gender categories, providing insights into various skin cancer conditions. The distri-

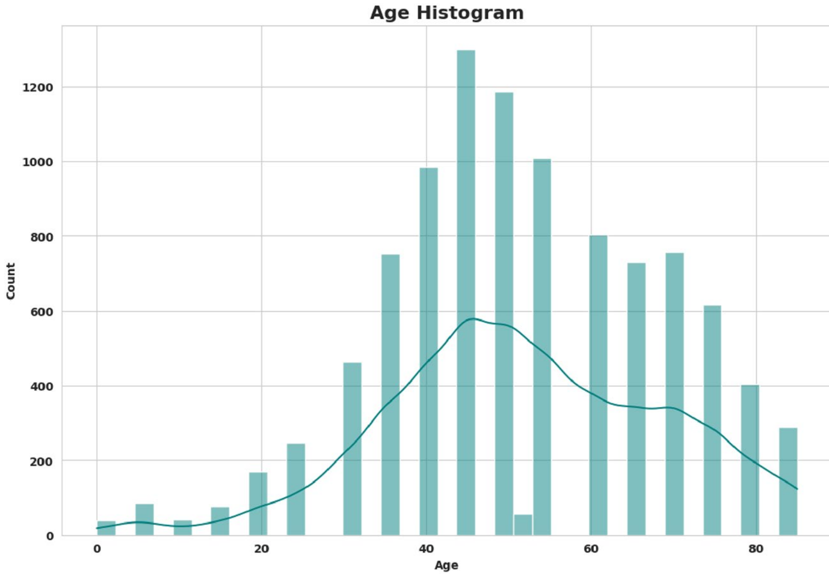
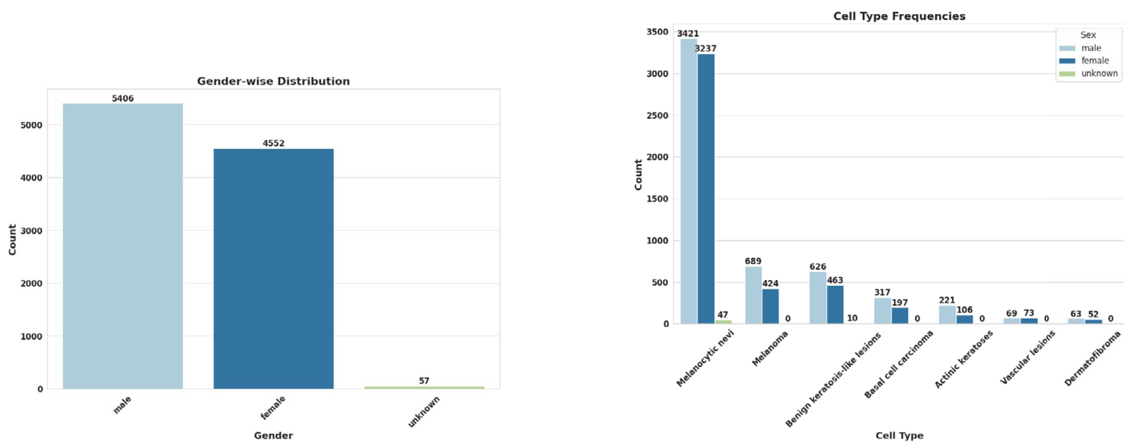


Fig. 3. Age distribution of skin cancer patients of the Employed HAM10000 dataset.



(a) Gender Distribution of Skin Cancer Patients.

(b) Gender-wise Distribution of Skin Cancer Types.

Fig. 4. Distribution of the Employed HAM10000 dataset.

bution reveals a clear gender disparity, with males exhibiting a higher frequency than females. Melanocytic nevi have the highest number of cases, with 3,421 in males and 3,237 in females, while 47 cases fall under the unknown category. Melanoma is more prevalent in males (689) than in females (424). A similar pattern is observed for benign keratosis-like lesions, with 626 cases in males, 463 in females, and 10 in the unknown category. Basal cell carcinoma and actinic keratoses occur at moderate rates, with basal cell carcinoma affecting 317 males and 197 females, while actinic keratoses are recorded in 221 males and 106 females. Vascular lesions show a slight female predominance, with 69 cases in males and 73 in females. Lastly, dermatofibroma has minimal representation, with 63 cases in males and 52 in females.

After the preprocessing step, label encoding was performed to convert categorical labels into numerical values. These numerical labels were then transformed into a one-hot encoded format with unique identifiers for multiclass classification. The dataset was split into training and testing subsets in an 80:20 ratio using stratified sampling to preserve balanced distributions. This splitting ensured that both subsets maintained a representative class distribution. Table 1 presents the distribution of skin cancer types in the HAM10000 dataset, showing the number of training and test images before augmentation. Melanocytic nevi constitute 66% of the images, while melanoma and benign keratosis-like lesions account for 11.11% and 10.97%, respectively. Other skin cancer types have a relatively low representation. To address this imbalance, conventional image augmentation techniques were applied through oversampling. These techniques include adjustments in brightness, contrast,

Cancer Types	Training Images (Before Augmentation)	Training Images (After Augmentation)	Test Images
Melanocytic nevi	5364	10000	1341
Melanoma	890	10000	223
Benign keratosis-like lesions	879	10000	220
Basal cell carcinoma	411	10000	103
Actinic keratoses	262	10000	65
Vascular lesions	114	10000	28
Dermatofibroma	92	10000	23

Table 1. Training and test images of skin cancer types.

left-right flipping, hue, and saturation, among others. The number of instances for each skin cancer type was increased and capped at 10,000 to achieve a more balanced dataset.

Multi-stage knowledge distillation

Knowledge distillation is an optimization technique that transfers knowledge from larger models to smaller ones[37]. The primary objective of this method is to train a compact model to replicate the behavior of a larger, more complex model. Top-performing classification models are often large, computationally expensive, and slow. Knowledge distillation addresses these limitations by producing a distilled model that is relatively small, faster, and more efficient. The distilled model retains comparable performance while requiring fewer trainable parameters. The knowledge distillation framework is not limited to a single neural network architecture; it consists of a teacher-student model paradigm. Additional models can further enhance the distillation process[38]. The teacher model is pre-trained and generates soft targets, which are probability distributions predicted by the large model[39]. The student model learns from these soft targets in conjunction with ground truth labels. The Kullback-Leibler (KL) divergence loss function is applied between the teacher and student outputs, while cross-entropy loss is used between the ground truth and student outputs. A balancing factor, alpha (α), regulates the contribution of these loss functions. Additionally, a temperature hyperparameter is introduced to scale the softmax function, thereby smoothing the probability distributions[40].

The distinct types of knowledge distillation have been introduced in recent years. This study establishes the multi-stage knowledge distillation technique based on features with intermediate exits. It is very similar to the feature-based distillation techniques. The knowledge distillation progresses through multiple stages. An additional intermediate loss function with two balancing factors, defined as beta (β) and gamma (γ), is incorporated into this procedure. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{intermediate}} + \beta \cdot \mathcal{L}_{\text{logits}} + \gamma \cdot \mathcal{L}_{\text{classification}} \tag{1}$$

Total loss combines three different loss components: classification loss, logit distillation loss, and intermediate loss. These losses were weighted by the coefficients of the balancing factors, i.e., α, β, and γ. The parameters α, β, and γ were automatically optimized using the Optuna framework with the constraint $\alpha + \beta + \gamma = 1$. This optimization was applied consistently across both single-stage and multi-stage knowledge distillation setups.

From an optimization perspective, multi-stage knowledge distillation introduces hierarchical supervision at different depths of the student network. This approach improves gradient flow and allows the lightweight student model to learn multi-scale representations that align with the teacher model's semantic hierarchy. Each intermediate exit provides additional gradients, effectively regularizing training and reducing overfitting. This step leads to a better generalization bound compared to traditional single-stage distillation.

- **Intermediate loss**: The intermediate loss $\mathcal{L}_{\text{intermediate}}$ is the mean of intermediate losses between the teacher's intermediate outputs and the student's intermediate outputs. It is mathematically represented as:

$$\mathcal{L}_{\text{intermediate}} = \frac{(\mathcal{L}_{\text{classification}}(T_0, S_0) + \mathcal{L}_{\text{classification}}(T_1, S_1))}{2.0} \tag{2}$$

Where T_0 and T_1 are intermediate outputs of the teacher model with the softmax activation function, S_0 and S_1 are intermediate outputs of the student model with the softmax activation function, and $\mathcal{L}_{\text{classification}}$ represents the classification loss function, the categorical cross-entropy function used in this study.

- **Distillation loss**: The distillation loss, denoted as $\mathcal{L}_{\text{logits}}$, represents the loss transferred when the student model's final logits (S_2) to mimic the teacher model's final logits (T_2). This process uses a temperature parameter (T_{temp}) to smooth the logits, facilitating the transfer of soft knowledge. The equation is defined as:

$$\mathcal{L}_{\text{logits}} = \mathcal{L}_{\text{distill}}(T_2, S_2, T_{\text{temp}}) \tag{3}$$

where L_{distill} is defined as:

$$\mathcal{L}_{\text{distill}}(T_2, S_2, T_{\text{temp}}) = \frac{1}{N}\sum_{i=1}^{N} \text{KL}\left(P^{(i)}_{\text{teacher}} \parallel P^{(i)}_{\text{student}}\right) \tag{4}$$

In (3) and (4), T_2 is the Logits from the output layer of the teacher model, S_2 is the logits from the output layer of the student model, T_{temp} the temperature hyperparameter is used to smooth the logits from harder to softer probabilities, and KL Divergence is the measure of the difference of distributions between the teacher's and student's output, $P^{(i)}_{\text{teacher}}$ is the softened probability distribution of the i^{th} example (teacher model), computed as:

$$P^{(i)}_{\text{teacher}} = \frac{\exp(T^{(i)}_{-1}/T_{\text{temp}})}{\sum_j \exp(T^{(j)}_{-1}/T_{\text{temp}})}$$

- $P^{(i)}_{\text{student}}$ is the softened probability distribution of the i-th example (Student model), computed as:

$$P^{(i)}_{\text{student}} = \frac{\exp(T^{(i)}_{-1}/T_{\text{temp}})}{\sum_j \exp(T^{(j)}_{-1}/T_{\text{temp}})}$$

- **Classification loss**: $\mathcal{L}_{\text{classification}}$ represents the classification loss function computed by cross-entropy, which measures the difference between the ground truth y_{true} and the predicted logits from the student model S_2.

$$\mathcal{L}_{\text{classification}} = \mathcal{L}_{\text{classification}}(y_{\text{true}}, S_2) \tag{5}$$

Fig. 5 illustrates the proposed multi-stage knowledge distillation framework for skin lesion diagnosis applied in this research. The process consists of two primary components: a hybrid teacher network and a hybrid student framework. The teacher ensemble model integrates ViT and ConvNeXT deep learning architectures, while the student ensemble model combines CNN and EfficientNet. Multiple intermediate exits are present in these hybrid models, allowing for early predictions. The intermediate loss is computed by averaging the loss values across these exits. The distillation loss is measured between the output layers of the teacher and student networks. The categorical cross-entropy loss is also applied to compare the student model's predictions with the ground truth labels.

The teacher and student models process input images of size 64 $\times$ 64. The ViT architecture utilizes Multi-Head Attention, Patch Embedding, and Transformer encoder blocks for feature extraction. ConvNeXT is structured with DepthwiseConv2D layers and batch normalization. The output layers and intermediate exits are derived through global pooling and dense layers. The extracted features from both architectures are fused into a single feature vector.

The CNN model consists of Conv2D, MaxPooling2D, and flattening layers, while EfficientNet concludes with global average pooling and dense layers. Dropout and regularization techniques are incorporated into the fusion layers to enhance generalization. The output layers from both architectures are concatenated into a single feature vector for classification.

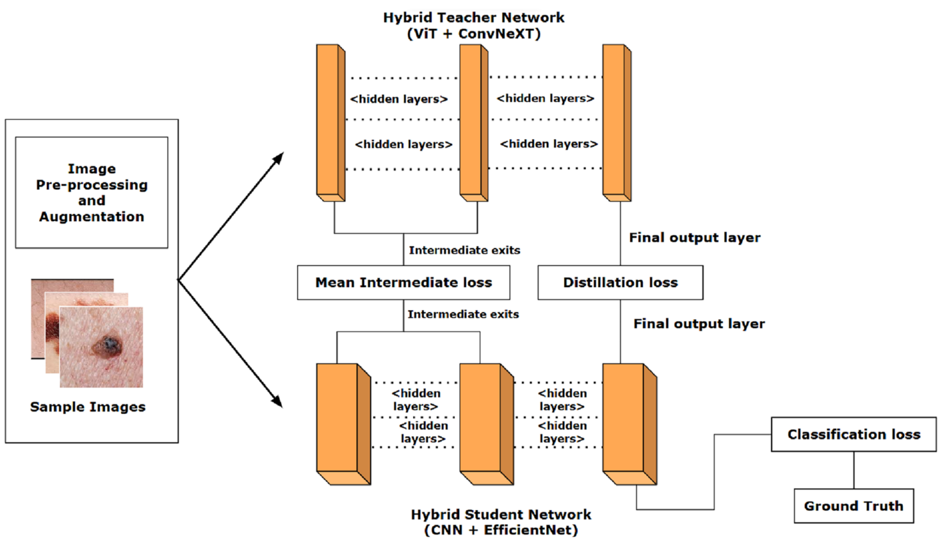


Fig. 5. Proposed multi-stage knowledge distillation framework used in this work.

Layer (Type)	Output Shape	Params	Connected to
`InputLayer`	(None, 64, 64, 3)	0	-
`Conv2D`	(None, 64, 64, 16)	448	InputLayer
`MaxPool`	(None, 32, 32, 16)	0	Conv2D
`Conv2D`	(None, 32, 32, 32)	4,640	MaxPool
`MaxPool`	(None, 16, 16, 32)	0	Conv2D
`EfficientNetB0`	(None, 2, 2, 1280)	4,049,571	InputLayer
`Flatten`	(None, 8192)	0	MaxPool
`GAP`	(None, 1280)	0	EfficientNetB0
`Dense (Exit 1)`	(None, 7)	57,351	Flatten
`Dense (Exit 2)`	(None, 7)	8,967	GAP
`Concat`	(None, 14)	0	Exit 1, Exit 2
`Dense`	(None, 256)	3,840	Concat
`Dropout`	(None, 256)	0	Dense
`Dense (Final)`	(None, 7)	1,799	Dropout

Table 2. Proposed hybrid student architecture.

Block / Layer	Output Shape	Param#	Notes
Input Layer	(64, 64, 3)	0	Raw image input
Patch Embedding + Encoder	(256, 256)	78K	Linear patch projection + position embedding
Transformer Block $\times$ 2	(256, 256)	$\sim$2.1M each	MHA + FFN with residuals
Conv Block $\times$ 2	(16, 16, 256)	$\sim$80K each	Conv2D + Depthwise + BN + Add
Global Pooling (1D + 2D)	(256 + 256)	0	Parallel pooling branches
Concatenate + Dense (512)	(512)	262K	Feature fusion
Output Heads	(7 classes)	$\sim$1.8K each	ViT exits, Conv exits, Combined output

Table 3. Proposed hybrid teacher architecture.

Mathematically, let f_i be feature maps from intermediate layers. The fused representation $F = \sum_{i=1}^{n} w_i \cdot f_i$ captures diverse abstraction levels, where w_i are trainable weights. This fusion enhances representational richness and injects complementary semantics, improving the student model's capacity to approximate the teacher's decision boundary with fewer parameters.

In the proposed hybrid teacher (demonstrated in Table 3) and student architecture, the concatenation-based layer fusion technique is applied. For the teacher network, the ViT model produces a feature embedding and the ConvNeXT branch outputs as $f_{ViT} \epsilon \mathbb{R}^{d_v}$ and $f_{Conv} \epsilon \mathbb{R}^{d_c}$.

These embeddings are fused via concatenation:

$$f_{\text{fusion}} = [f_{\text{ViT}}; f_{\text{Conv}}] \in \mathbb{R}^{d_v + d_c}. \tag{6}$$

This fused vector is then projected through a dense layer to produce the final prediction:

$$y_{\text{teacher}} = \text{softmax}(W_{\text{fusion}} f_{\text{fusion}} + b_{\text{fusion}}). \tag{7}$$

In the student model, fusion occurs at the decision level. The softmax outputs from the CNN and EfficientNet branches are concatenated as:

$$p_{\text{fusion}} = [p_{\text{CNN}}; p_{\text{Eff}}] \in \mathbb{R}^{2C}. \tag{8}$$

Finally, a dense layer projects the corresponding fused decision vector into the class space:

$$y_{\text{student}} = \text{softmax}(W_{\text{student}} p_{\text{fusion}} + b_{\text{student}}). \tag{9}$$

Supporting layers such as **GlobalAveragePooling2D**, **MaxPooling2D**, **Flatten**, **BatchNormalization**, and **Rescaling** are applied before fusion to prepare feature vectors of compatible shapes and scales. The combined complementary features, i.e., global context from ViT and local patterns from ConvNeXT and CNN, increase representational capacity, reduce branch-specific errors, and improve overall classification performance.

Table 4 summarizes the total number of parameters for the hybrid models, including both trainable and non-trainable parameters. The hybrid teacher model comprises a size of 20.03 MB with 5,249,073 and 6 kB trainable and non-trainable parameters, respectively. The hybrid student model, demonstrated in Table 2, has a size of 4.02 MB and includes 1,054,695 trainable parameters. The proposed multi-stage knowledge distillation is illustrated

Model	Total	Trainable	Non-Trainable
Teacher (ViT + ConvNeXT)	5,250,609	5,249,073	1,536
Size	20.03 MB	20.02 MB	6.00 kB
Student (CNN + EfficientNet)	1,054,695	1,054,695	0
Size	4.02 MB	4.02 MB	0.00 B

Table 4. Parameter summary of the hybrid teacher and student models.

in Algorithm 1. This model begins by initializing hybrid models and tunable hyperparameters. While training, the teacher and student models produce outputs, which measure total loss. The total and other loss functions are also defined to execute the operation. Gradients and optimizers are computed and updated during each epoch over mini-batches. The proposed hybrid student network (CNN and EfficientNet) uses CNN's locality-aware inductive bias and EfficientNet's compound scaling, enabling efficient feature extraction with fewer parameters. Compared to transformer-only models that rely heavily on global attention, the hybrid structure performs better under data imbalance and resource constraints.

Data: Training and Validation data, Teacher model, Student model
Result: Distilled model
$teacher\_model \leftarrow$ HybridTeacherModel (ViT + ConvNeXT) ;
$student\_model \leftarrow$ HybridStudentModel (CNN + EfficientNet) ;
$\alpha \leftarrow FeatureLoss$, $\beta \leftarrow LogitsLoss$, $\gamma \leftarrow ClassLoss$; $T \leftarrow Temperature$;

Function ClassificationLoss(y_{true}, y_{pred}): ;
  return categorical_crossentropy(y_{true}, y_{pred}) ;

Function KLDivergence(P_t, P_s): ;
  $div \leftarrow 0$;
  for *each i in* P_t *and* P_s **do**
    $div \leftarrow div + P_t[i] \cdot \log \frac{P_t[i]}{P_s[i]}$;
  end
  return div ;

Function IntermediateLoss(T, S): ;
  $loss \leftarrow 0$;
  for *each pair* (T_i, S_i) *in* T, S **do**
    $loss \leftarrow loss+$ ClassificationLoss(T_i, S_i) ;
  end
  return $loss/(\#\ pairs)$;

Function DistillationLoss($logits_t, logits_s, temperature$): ;
  $P_t \leftarrow softmax(logits_t/temperature)$;
  $P_s \leftarrow softmax(logits_s/temperature)$;
  return KLDivergence(P_t, P_s) ;

Function TotalLoss($outputs_t, outputs_s, T, S, y$): ;
  $L_{int} \leftarrow$ IntermediateLoss(T, S) ;
  $L_{logit} \leftarrow$ DistillationLoss($outputs_t, outputs_s, temperature$) ;
  $L_{class} \leftarrow$ ClassificationLoss($y, outputs_s$) ;
  return $\alpha L_{int} + \beta L_{logit} + \gamma L_{class}$;

for *epoch = 1* **to** *epochs* **do**
  for *each batch* (x, y) *in data_loader* **do**
    $(T, outputs_t) \leftarrow teacher\_model(x)$;
    $(S, outputs_s) \leftarrow student\_model(x)$;
    $loss \leftarrow$ TotalLoss($outputs_t, outputs_s, T, S, y$) ;
    optimizer.zero_grad() ;
    loss.backward() ;
    optimizer.step() ;
  end
end

Algorithm 1. Proposed multi-stage knowledge distillation model.

Explainable AI methods

Explainable Artificial Intelligence (XAI) focuses on making AI models and their decisions understandable and interpretable to humans. It ensures that decisions are transparent and bias-free while identifying potential model failures. XAI comprises various applications across domains such as healthcare diagnosis, finance, law, and autonomous systems. It explains decision-making by demonstrating how changes in input can influence outcomes. The employed XAI methods in this study are briefly discussed below.

1. Grad-CAM refers to "Gradient-weighted Class Activation Mapping" " that highlights the important regions of a single image[41]. It uses gradients of the target class. Averaging the gradients shows the importance of each feature map. After weighted feature map computation, a generated heatmap was experimented with various important regions. It relies heavily on gradients, so the problem of exploding or vanishing gradients might be problematic.
2. Score-CAM refers to "Score-weighted Class Activation Mapping" an improved version of Grad-CAM that reduces the dependency on gradients[42]. It perturbs the input image by masking it with feature maps. After perturbing the image, the model's output score measures the significance of the feature map. A generated heatmap is experimented with by combining these important scores. It does not depend on gradients, avoiding robust issues, e.g., exploding or vanishing gradients. However, it is computationally expensive because it requires multiple passes through the applied classification approach.
3. LIME: Local Interpretable Model-agnostic Explanations (LIME) explains predictions of any black-box model that are at least interpretable[43]. It works by perturbing the input image and randomly masking parts of the image. It also observes the model's prediction change in each perturbation. It fits a simple interpretable model to approximate the relationship between input features and output predictions. This XAI technique can work with various data types, including images, text, tabular, and models. However, it can be computationally expensive to reason about repetitive perturbations. It also provides local explanations rather than global ones for the model.

Results and discussion

Experimental setup

This section outlines the experimental setup, including key parameters used for the applied baseline models and knowledge distillation techniques to achieve optimal performance.

Table 5 lists the optimized hyperparameters of the applied models, automatically determined using the Optuna framework. The temperature value is set to 10 to produce softer target distributions. The weighted components alpha and beta are set to 0.25, facilitating improved feature learning through intermediate representations of the teacher model while capturing richer information beyond ground truth from the teacher's softened predictions. The gamma value is set to 0.5 to focus on learning from actual ground truth for classification. In the multi-stage knowledge distillation setup, loss weights are equally weighted (50%) for distillation and classification. The balancing factors differ for single-stage knowledge distillation, where the alpha value is deactivated to freeze intermediate feature learning. In contrast, beta and gamma values are set to 0.5 to ensure balanced learning from both sources.

Results of the baseline models

The baseline models, including transformer-based and fine-tuned models, regarding resource efficiency and performance across key metrics, are analyzed in this section. These models were trained on imbalanced data without augmentation. Table 6 presents the performance metrics of all baseline models implemented in this study. EfficientNet achieved the highest accuracy of 0.7794, followed by ConvNeXT with 0.7529. CNN, DenseNet, ViT, and EANet demonstrated moderate performance, with accuracy values of 0.7382, 0.7218, 0.7265, and 0.7190, respectively. ConvNeXT achieved the highest precision of 0.8370, while CNN performed well at 0.8147. EfficientNet, DenseNet, ViT, EANet, and ResNet recorded precision values of 0.8027, 0.8037, 0.8004, 0.7971, and 0.7747, respectively. EfficientNet attained the highest recall of 0.7573, while ConvNeXT, EANet, ViT, DenseNet, and CNN achieved recall values of 0.6971, 0.6566, 0.6599, 0.6565, and 0.6814, respectively, demonstrating strong performance. ResNet had the lowest recall at 0.5966. The F1-score of ResNet was also

Tunable Hyperparameters	
Common Hyperparameters	
Learning Rate	0.001, $\min\_\mathrm{lr} = 10^{-6}$
Optimizer	Adam
Batch Size	32
Epochs	100
ReduceLROnPlateau	$\mathrm{factor} = 0.2, \mathrm{patience} = 3$
Early Stopping	$\mathrm{patience} = 5$
Knowledge Distillation Hyperparameters	
Temperature (T)	10.0
Intermediate Feature Loss (α)	0.25 (N/A*)
Logits Loss (β)	0.25 (0.5*)
Final Classification Loss (γ)	0.4 (0.5*)

Table 5. Tunable hyperparameters for training and multi-stage knowledge distillation (* denotes values for single-stage knowledge distillation).

Model	Accuracy	Precision	Recall	F1-score	AUC
CNN	0.7382	0.8147	0.6814	0.7420	0.9592
ResNet	0.6607	0.7747	0.5966	0.6741	0.9195
EfficientNet	**0.7794**	0.8027	0.7573	0.7793	0.9646
DenseNet	0.7218	0.8037	0.6565	0.7226	0.9533
ViT	0.7265	0.8004	0.6599	0.7233	0.9580
EANet	0.7190	0.7971	0.6566	0.7200	0.9579
MobileViT	0.6949	0.7656	0.6311	0.6920	0.9082
ConvNeXT	0.7529	**0.8370**	**0.6971**	**0.7606**	**0.9660**

Table 6. Baseline model performance metrics.

Baseline Model	Epoch elapsed	Loss	Training (ms/step)	Inference (ms/step)
CNN	23	0.6950	4	2
ResNet	62	0.9656	14	12
EfficientNet	14	0.6902	75	19
DenseNet	19	0.7444	19	15
ViT	12	0.7016	67	22
EANet	17	0.7072	111	40
ConvNeXT	33	0.6360	9	4

Table 7. Benchmarking baseline models on resource consumption.

the lowest at 0.6741, whereas CNN, EfficientNet, DenseNet, ViT, EANet, and ConvNeXT recorded F1-scores of 0.7420, 0.7793, 0.7226, 0.7233, 0.7200, and 0.7606, respectively. EfficientNet and ConvNeXT exhibited the highest F1-scores among the baseline models. All baseline models achieved AUC scores above 90%. CNN and EfficientNet had the highest AUC scores of 0.9592 and 0.9646, respectively, making them suitable candidates for the hybrid student network. ViT and ConvNeXT recorded AUC scores of 0.9580 and 0.9660, respectively, which led to their selection as the transformer models for the hybrid teacher network. The multi-stage knowledge distillation framework was constructed based on these hybrid models. Other models, such as ResNet, DenseNet, and EANet, achieved AUC scores of 0.9195, 0.9533, and 0.9579, respectively. Due to resource constraints, these models were not incorporated into the multi-stage knowledge distillation process.

Table 7 presents the resource consumption metrics, including loss values, epochs elapsed, training time per step, inference time, and memory consumption for the baseline models. The experiments were conducted on the Kaggle platform using a P100 GPU. ConvNeXT demonstrated the lowest loss of 0.6360 after 33 epochs, with a training time of 9 ms/step and an inference time of 4 ms/step. EfficientNet performed comparably, achieving a loss of 0.6902 in just 14 epochs, though it required a significantly higher training time of 75 ms/step and inference time of 19 ms/step. CNN, the simplest model among those tested, reached a loss of 0.6950 in 23 epochs, with a fast training time of 4 ms/step and inference time of 2 ms/step. ResNet required a longer training duration of 62 epochs, resulting in a relatively high loss of 0.9656, with a training time of 14 ms/step and an inference time of 12 ms/step. DenseNet completed training in 19 epochs, achieving a loss of 0.7444, with a training time of 19 ms/step and inference time of 15 ms/step. ViT, a transformer-based model, converged in 12 epochs with a moderate loss of 0.7016 but required a higher training time of 67 ms/step and inference time of 22 ms/step. EANet, completing training in 17 epochs, recorded a loss of 0.7072 but was computationally expensive, requiring 111 ms/step for training and 40 ms/step for inference. ConvNeXT proved to be the most efficient among all models, while CNN and its variants demonstrated strong efficiency in training and inference time.

Comparative analysis of hybrid student networks, single-stage knowledge distillation, multi-stage knowledge distillation, and adaptive layer fusion enhancements

In this work, ViT and ConvNeXT are chosen for the proposed hybrid transformer teacher model due to their superior performance, with ConvNeXT achieving the best overall performance while maintaining reasonable resource consumption. In contrast, CNN and EfficientNet are selected for the hybrid student model since CNN is computationally efficient with a low training and inference time. EfficientNet balances strong accuracy with relatively lower training time than other high-performing models. A detailed comparative analysis of the performance and effectiveness of various methods, including the hybrid student baseline network, multi-stage knowledge distillation, and the impact of layer fusion enhancements on model optimization, is presented in the following paragraphs. Table 8 summarizes various performance metrics with 5-fold cross-validation of the final output, intermediate exits, and loss values for the methods evaluated in this study. These methods include the hybrid student baseline, multi-stage knowledge distillation, and multi-stage knowledge distillation with layer fusion. The hybrid baseline student model recorded a loss value of 2.7705 ± 0.0018, demonstrating a relatively high training error when compared to evolved methods. For intermediate predictions, the categorical cross-

Architecture	Loss	CC_1	CC_2	CH_1	CH_2
Hybrid					
Student	2.7705	0.4091	1.9817	0.3482	1.0603
Baseline	± 0.0018	± 0.0012	± 0.0014	± 0.0015	± 0.0017
Single-Stage	0.1949	N/A	N/A	N/A	N/A
Distillation	± 0.0004				
Multi-Stage	0.6873	2.4205	1.9465	1.1619	1.0102
Distillation	± 0.0006	± 0.0011	± 0.0010	± 0.0005	± 0.0006
Multi-Stage					
Distillation +	0.6329	2.1940	2.1279	1.2746	1.0272
Fused layer	± 0.0003	± 0.0013	± 0.0006	± 0.0003	± 0.0009
Architecture	**Accuracy**	**Precision**	**Recall**	**F1-score**	**AUC**
Hybrid					
Student	0.8741	0.8988	0.8534	0.8757	0.9849
Baseline	± 0.0010	± 0.0021	± 0.0013	± 0.0023	± 0.0006
Single-Stage	0.9002	0.9125	0.8873	0.9000	0.9857
Distillation	± 0.0009	± 0.0008	± 0.0004	± 0.0007	± 0.0002
Multi-Stage	0.9223	0.9303	0.9159	0.9233	0.9879
Distillation	± 0.0006	± 0.0010	± 0.0009	± 0.0016	± 0.0003
Multi-Stage					
Distillation +	0.9606↑	0.9622↑	0.9577↑	0.9591↑	0.9905↑
Fused layer	± 0.0043	± 0.0028	± 0.0010	± 0.0002	± 0.0009

Table 8. Performance comparison of knowledge distillation techniques with 5-fold cross-validation for skin lesion classification.

entropy (CC) scores were 0.4091 ± 0.0012 for the first exit (CC_1) and 1.9817 ± 0.0014 for the second exit (CC_2), while the categorical hinge (CH) scores were 0.3482 ± 0.0015 for the first exit (CH_1) and 1.0603 ± 0.0017 for the second exit (CH_2). These results imply that the first intermediate exit performed satisfactorily in early-stage predictions, demonstrating lower loss values and more dedicated results compared to the second exit. The second exit, in contrast, displayed moderate performance with higher losses, suggesting that deeper layers added complicatedness without proportional improvement in intermediate output accuracy. With the application of multi-stage knowledge distillation, the model's loss significantly decreased to 0.6873 ± 0.0006, demonstrating the effectiveness of transferring knowledge from teacher to student in evolved stages. Combining layer fusion into this process further lessened the loss to 0.6329 ± 0.0003, revealing a meaningful enhancement in optimization and training stability. Regarding intermediate metrics, the initial multi-stage model reported CC_1 and CH_1 values of 2.4205 ± 0.0011 and 1.1619 ± 0.0005, respectively, while CC_2 and CH_2 were 1.9465 ± 0.0010 and 1.0102 ± 0.0006. Upon incorporating layer fusion, CC_1, and CH_1 were improved to 2.1940 ± 0.0013 and 1.2746 ± 0.0003, respectively. Similarly, CC_2 and CH_2 slightly modified to 2.1279 ± 0.0006 and 1.0272 ± 0.0009. Despite a slight decline in the first intermediate exit performance, which may be attributed to added complexity in a resource-constrained environment, the overall enhancement in final prediction quality confirms the effectiveness of the fusion mechanism. The single-stage knowledge distillation method yielded the most down loss value of 0.1949 ± 0.0004, showing adequate training convergence. However, this approach did not include intermediate exits; thus, categorical cross-entropy and hinge scores were not usable. Despite this limitation, the model indicated competitive results when assessing end-to-end performance. In terms of classification metrics, the hybrid baseline student model earned an accuracy of 87.41% $\pm 0.10\%$, a precision of 89.88% $\pm 0.21\%$, a recall of 85.34% $\pm 0.13\%$, an F1-score of 87.57% $\pm 0.23\%$, and an AUC of 98.49% $\pm 0.06\%$. These results provide a strong baseline but leave space for improvement. The single-stage knowledge distillation model beat the baseline, achieving an accuracy of 90.02% $\pm 0.09\%$, precision of 91.25% $\pm 0.08\%$, recall of 88.73% $\pm 0.04\%$, F1-score of 90.00% $\pm 0.07\%$, and AUC of 98.57% $\pm 0.02\%$, thus demonstrating constant improvements across most evaluation benchmarks. The multi-stage knowledge distillation approach further improved the performance, with an accuracy of 92.23% $\pm 0.06\%$, precision of 93.03% $\pm 0.10\%$, recall of 91.59% $\pm 0.09\%$, F1-score of 92.33% $\pm 0.16\%$, and AUC of 98.79% $\pm 0.03\%$. These gains confirm the advantage of distributing knowledge transfer across multiple stages. Most notably, the integration of layer fusion with multi-stage distillation produced the best outcomes across all metrics: accuracy of 96.06% $\pm 0.43\%$, precision of 96.22% $\pm 0.28\%$, recall of 95.77% $\pm 0.10\%$, F1-score of 95.91% $\pm 0.02\%$, and AUC of 99.05% $\pm 0.09\%$, representing a considerable enhancement, with all performance metrics exceeding 95%, marking this configuration as the most influential in the study. Fig. 6 illustrates the variation of intermediate exit metrics, e.g., categorical cross-entropy and categorical hinge, with the change of epochs of the final optimized, distilled model. There are sudden spikes in the green line, corresponding to the second intermediate exits of categorical cross-entropy for training, showing instability during training. However, other lines or exit metrics remain low during the training phase. Fig. 7 presents the final output metrics and loss trends versus epochs for the final model

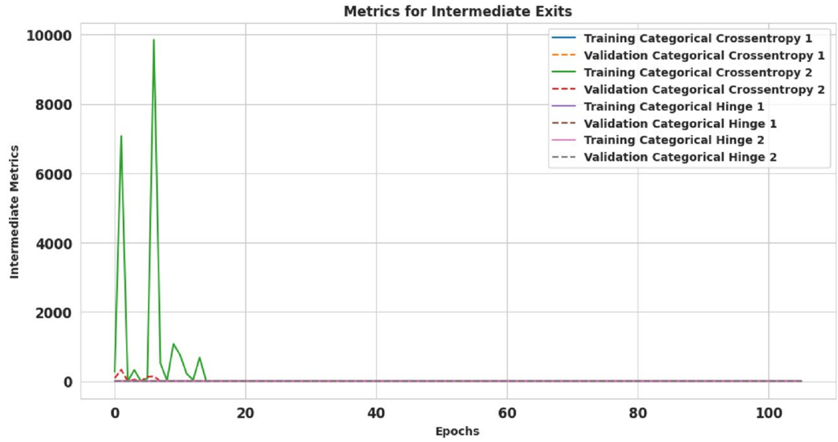


Fig. 6. Evaluation of multi-stage knowledge distillation: intermediate exit metrics.

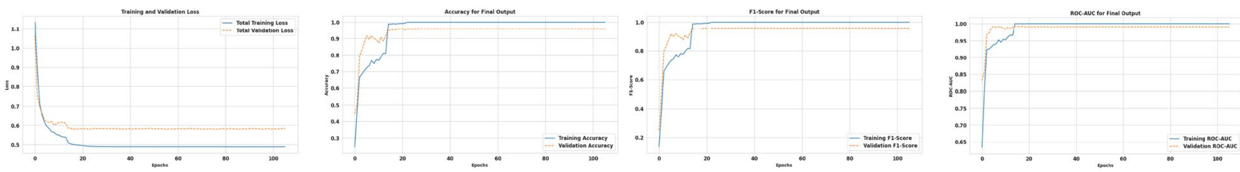


Fig. 7. Evaluation of multi-stage knowledge distillation: loss trends, and performance metrics (Accuracy, F1-score, and ROC-AUC).

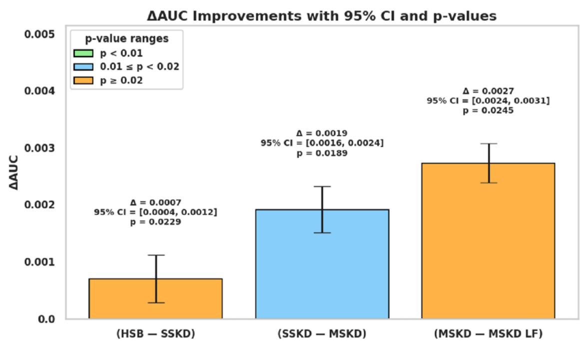


Fig. 8. Comparative performance analysis of the hybrid baseline, single-stage, and multi-stage knowledge distillation models (with and without layer fusion).

incorporating layer fusion. The loss curves exhibited a rapid decline within the first 10 epochs. Between epochs 10 and 20, the rate of decrease slowed, and the validation loss stabilized. The model appears to have converged effectively, reaching minimal loss values with no significant signs of overfitting. Several metric curves are depicted as subfigures, illustrating training and validation trends. These curves followed similar and consistent paths for training and validation lines. After 10–20 epochs, validation metrics reached approximately 90–95%, while training metrics approached approximately 100%. An insignificant gap is observed across all metrics, indicating slight overfitting. However, the model retained strong generalization performance on the test data. Figure 8 demonstrates the progress in ΔAUC across subsequent stages of the knowledge distillation framework, accompanied by 95% confidence intervals and p-values that assess the statistical significance of these refinements. The first comparison, between the Hybrid Student Baseline (HSB) and Single-Stage Knowledge Distillation (SSKD), delivers a modest AUC gain of 0.0007 with a 95% confidence interval of [0.0004, 0.0012] and a p-value of 0.0229, indicating a statistically significant yet relatively smallish improvement, meaning that even a single-stage of distillation can enhance model performance. The second comparison, between SSKD and Multi-Stage Knowledge Distillation (MSKD), reveals a more major progress of 0.0019 in AUC, with a narrower 95% confidence interval of [0.0016, 0.0024] and a p-value of 0.0189, demonstrating that rising the distillation process across multiple stages yields more substantial performance gains, aiding from deeper features and more gradual knowledge transfer from the teacher to the student model. Finally, the transition from MSKD to MSKD with

Modified Architecture	Epoch elapsed	Training (ms/step)	Inference (ms/step)
Hybrid Student			
Baseline	16	240	40
Multi-Stage Distillation	32	79	16
Single-Stage Distillation	52	79	16
Multi-Stage Distillation + Layer Fusion	57	61	15

Table 9. Computational efficiency of knowledge distillation models on resource consumption.

Dominant Emphasis	Intermediate Loss (α)	Logit Distillation Loss (β)	Classification Loss (γ)	Logical Constraint
Balanced Baseline	0.25	0.25	0.5	$\alpha + \beta = \gamma$
Distillation Loss Priority	0.3	0.3	0.4	$\alpha + \beta > \gamma$
Classification Loss Priority	0.2	0.2	0.6	$\alpha + \beta < \gamma$
Feature Representation Priority	0.5	0.2	0.3	$\alpha + \beta > \gamma,\ \alpha > \beta$
Logit Distillation Priority	0.2	0.5	0.3	$\alpha + \beta > \gamma,\ \alpha < \beta$

Table 10. Loss weight configuration and associated logical constraints by dominant emphasis.

Layer Fusion (MSKD LF) guides to the highest observed AUC improvement of 0.0027, with a 95% confidence interval of [0.0024, 0.0031] and a p-value of 0.0245. Although the p-value slightly surpasses the 0.02 threshold, the compatible upward direction and statistically significant gain certify the added value of incorporating layer fusion. Overall, Fig. 8 emphasizes a progressive enhancement in classification performance at each distillation stage, validating the efficacy of the proposed multi-stage and fusion-based knowledge distillation technique. Table 9 presents the benchmarking comparison of the optimized methods used in the study on the NVIDIA P100 GPU. The hybrid student baseline completes training on 16 epochs with 240 ms/step for training and 40 ms/step for inference. The single-stage knowledge distillation acquired a longer duration of 52 epochs but with a lower training time of 79 ms/step and 16 ms/step for inference. The multi-stage knowledge distillation attained a slightly shorter duration of 32 epochs but with a similar training time of 79 ms/step and 16 ms/step for inference. However, after adding layer fusion, it took the highest duration of 57 epochs with a lower training time of 61 ms/step and 15 ms/step.

Impact of logically-constrained loss weighting in multi-stage knowledge distillation

To clarify the role of supervisory focus within the training dynamics of hybrid transformer-based student networks, we conducted a comprehensive ablation study analyzing the influence of logically constrained loss weighting approaches on model implementation. Specifically, the multi-objective loss function utilized in our multi-stage distillation framework merges three components: intermediate feature alignment loss (α), logit-based distillation loss (β), and primary classification loss (γ). By systematically modulating the comparative contributions of these components under logically coherent constraints, we aimed to investigate the extent to which different supervisory regimes affect the effectiveness of knowledge transfer and generalization.

Table 10 summarizes the reckoned configurations, organized by their prior focus on loss intensity. The Balanced Baseline adheres to the constraint $\alpha + \beta = \gamma$, thereby preserving parity between distillation and classification goals. The weighting coefficient was defined as $\alpha = 0.25$, $\beta = 0.25$, and $\gamma = 0.5$. In contrast, Two unbalanced configurations enabled for comparison and test asymmetric supervisory regimes: the Distillation Loss Priority configuration meets $\alpha + \beta > \gamma$, suggesting a greater reliance on teacher guidance, whereas the Classification Loss Priority setting executes $\alpha + \beta < \gamma$, highlighting the task-specific objective. To further interrogate the internal structure of distillation signals, the distillation-prioritized control is bifurcated into Feature Representation Priority ($\alpha > \beta$) and Logit Distillation Priority ($\alpha < \beta$), separating the impacts of structural versus output-level teacher supervision. Table 11 delineates the relative implementation of distinct loss weighting systems developed via logical constraints within a multi-stage knowledge distillation framework. The configuration represented by $\alpha + \beta > \gamma,\ \alpha > \beta$, referred to as the Feature Representation Priority control, consistently reached the highest performance across all metrics. This setup gained an AUC of 0.9987, accuracy of 97.07%, precision of 97.47%, recall of 96.89%, and an F1-score of 97.18%, strongly implying that highlighting intermediate feature alignment yields a richer inductive bias. The enhanced semantic transfer enabled by feature-level supervision seems to support superior decision boundary learning and improved robustness in downstream classification. In comparison, the Logit Distillation Priority configuration ($\alpha + \beta > \gamma,\ \alpha < \beta$) also adhered to a distillation-prioritized paradigm but revealed marginally low performance, achieving an AUC

Logical Constraint	Accuracy	Precision	Recall	F1-score	AUC
$\alpha + \beta = \gamma$	0.9606	0.9622	0.9577	0.9591	0.9905
$\alpha + \beta > \gamma$	0.9665	0.9637	0.9600	0.9618	0.9945
$\alpha + \beta < \gamma$	0.9517	0.9542	0.9483	0.9512	0.9875
$\alpha + \beta > \gamma,\ \alpha > \beta$	0.9707	0.9747	0.9689	0.9718	0.9987
$\alpha + \beta > \gamma,\ \alpha < \beta$	0.9672	0.9676	0.9602	0.9639	0.9974

Table 11. Evaluation metrics associated with different logical constraints.

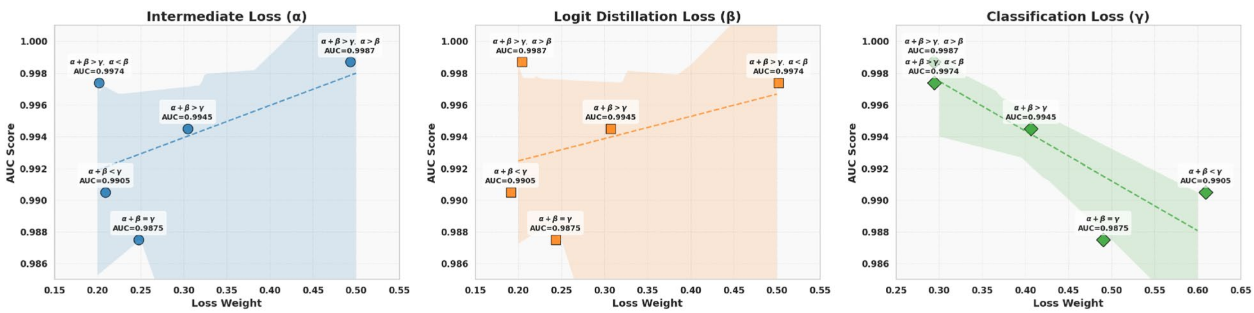


Fig. 9. AUC trends for varying loss weights under logical constraints.

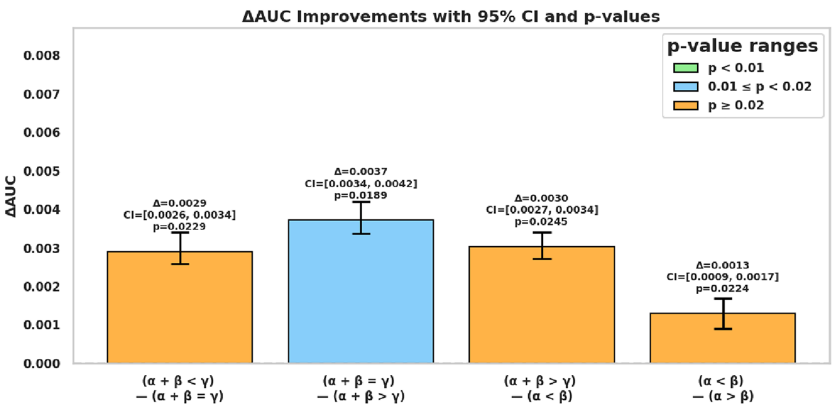


Fig. 10. ΔAUC improvements across key constraint comparisons with 95% confidence intervals and p-values.

of 0.9974, accuracy of 96.72%, precision of 96.76%, recall of 96.02%, and F1-score of 96.39%. Despite sharing the same overarching constraint ($\alpha + \beta > \gamma$), this performance gap highlights the crucial impact of interior loss composition, suggesting that feature-level guidance donates more meaningfully to semantic inference than output-level imitation independently. The Balanced Baseline condition, constrained by $\alpha + \beta = \gamma$, yielded sensible outcomes, including an AUC of 0.9905, accuracy of 96.06%, precision of 96.22%, recall of 95.77%, and F1-score of 95.91%, suggesting that equality between classification and distillation signals offers pleasing, though suboptimal, performance, potentially lacking the depth of semantic supervision fed by additional uneven configurations. Contrarily, the Classification Loss Priority control ($\alpha + \beta < \gamma$) displayed the lowest overall performance, with an AUC of 0.9875, accuracy of 95.17%, precision of 95.42%, recall of 94.83%, and F1-score of 95.12%, pronouncing decrease in performance proves that attenuating the distillation signal damages the model's ability to internalize the teacher's structural biases, resulting in feebler inference and reduced limitation bias. The general Distillation-Prioritized configuration ($\alpha + \beta > \gamma$) without extra compositional constraints delivered intermediate results, with an AUC of 0.9945, accuracy of 96.65%, precision of 96.37%, recall of 96.00%, and F1-score of 96.18%. These metrics validate the more general hypothesis that a strong distillation sign, regardless of its exact quota, is conducive to enhanced model fidelity. Collectively, these findings provide clear evidence that the structural logic underlying loss weighting is not only a hyperparameter tuning process but a foundational determinant of knowledge distillation effectiveness. Fig. 9 further supports this conclusion by depicting how AUC varies with differences in individual loss weights. A clear positive direction is observed for growing α, with the best AUC conducted under the feature representation priority configuration. While increasing β also

Optimization Technique	Accuracy	Precision	Recall	F1-score	AUC
Hybrid Student Baseline (HSB)	0.9222	0.9405	0.8855	0.9121	0.9860
Knowledge Distillation (KD)	0.8989	0.9543	0.8338	0.8899	0.9910
Post-Training Quantization (PTQ)	0.7813	0.8322	0.7548	0.7916	0.9693

Table 12. Evaluation metrics associated with fifferent size reduction methods for ISIC dataset.

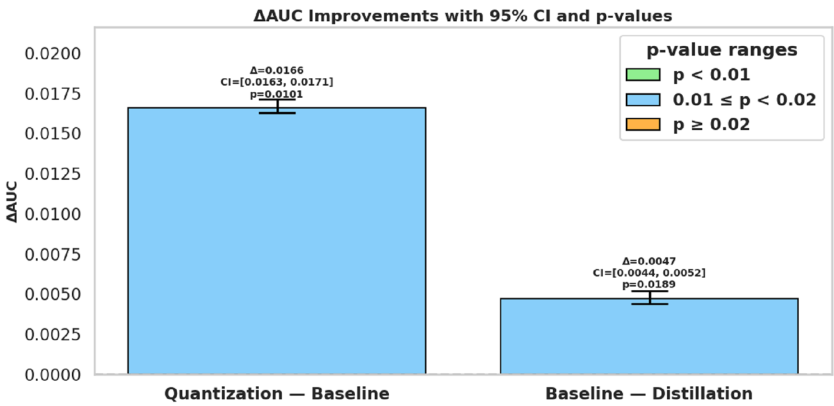


Fig. 11. ΔAUC improvements across size reduction methods with 95% confidence intervals and p-values.

improves AUC, the effect is more unpretentious. In contrast, increasing γ leads to an apparent decrease in AUC, revealing the negative impact of overemphasizing the classification purpose. These trends confirm that prioritizing intermediate feature alignment suggests the most effective semantic direction and that reduced dependence on classification loss improves inference within the distillation framework. Figure 10 presents Δ AUC comparisons with statistical testing. Notably, *Balanced vs. Classification* yielded ΔAUC = 0.0029 (CI = [0.0026, 0.0034], $p = 0.0229$); *Distillation-Prioritized vs. Balanced* yielded $\Delta = 0.0037$ (CI = [0.0034, 0.0042], $p = 0.0189$); and *Feature- vs. Logit-Prioritized* yielded $\Delta = 0.0013$ (CI = [0.0009, 0.0017], $p = 0.0224$)—all statistically significant.

Inference on ISISC 2018 test set

To assess the generalization performance of the presented optimization techniques, we executed inference on the ISISC 2018 test set, an established benchmark in information security imaging. We benchmarked three model configurations: the Hybrid Student Baseline (HSB), Knowledge Distillation (KD)-enhanced student, and the Post-Training Quantized (PTQ) model. Table 12 gives a comparative analysis of these methods using various classification metrics. The Hybrid Student Baseline (HSB) conducted the most congruous performance across all metrics, producing an accuracy of 92.22%, a precision of 94.05%, a recall of 88.55%, and an F1-score of 91.21%. Its AUC of 0.9860 further demonstrates strong prejudiced capability, suggesting a high degree of class separability. Conversely, the Knowledge Distillation (KD) technique attained an accuracy of 89.89%, a precision of 95.43%, a recall of 83.38%, and an F1-score of 88.89%. With an accuracy of 78.13%, a precision of 83.22%, and a recall of 75.48%, its F1-score of 79.16% and AUC of 0.9693, the Post-Training Quantization (PTQ) model illustrates satisfactory computational efficiency. Figure 11 presents the ΔAUC progress along with 95% confidence intervals (CIs) and associated p-values, color-coded according to statistical thresholds. Specifically, the (Quantization — Baseline) comparison produced a ΔAUC of 0.0166 (CI: [0.0163, 0.0171], $p = 0.0101$), revealing a statistically significant difference just past the $p < 0.01$ threshold. Meanwhile, the (Baseline — Distillation) contrast exhibited a more fair yet still statistically noteworthy ΔAUC of 0.0047 (CI: [0.0044, 0.0052], $p = 0.0189$).

Ablation study

Ablation studies have been performed to understand the impact of intermediate exits, layer fusion, and quantization on monitoring the performance of the applied techniques. Each ablation experiment modifies a specific component to analyze the benefits of the presence of the elements.

1. **Advantages of intermediate exits**: Table 13 compares single-stage and multi-stage knowledge distillation on intermediate exit presences, describing the progress of improvements in overall metrics. Although the multi-stage approach exhibits a higher loss (0.6873) compared to the single-stage method (0.1949), this is attributed to the inclusion of auxiliary losses for intermediate exits, which contribute to improved hierarchical feature learning. The multi-stage method achieves superior accuracy (92.23%) over the single-stage approach (90.02%), reflecting a +2.21% improvement in generalization. Precision, recall, and F1-score are also notably enhanced in the multi-stage approach, with improvements of +1.78%, +2.86%, and +2.33%,

Metric	Multi-Stage	Single-Stage	Progress	Possible Reason
Loss	0.6873	0.1949	**Higher loss**	Intermediate exits add auxiliary losses (CC1, CC2, CH1, CH2) for training.
Accuracy	0.9223	0.9002	+2.21%	Hierarchical feature learning from intermediate layers improves generalization.
Precision	0.9303	0.9125	+1.78%	Better discrimination of false positives through multi-scale feature extraction.
Recall	0.9159	0.8873	+2.86%	Enhanced sensitivity to true positives via intermediate supervision.
F1-score	0.9233	0.9000	+2.33%	Balanced precision-recall trade-off from staged learning.
AUC	0.9879	0.9857	+0.22%	Robust separability of classes due to multi-stage feature refinement.

Table 13. Advantages of intermediate exits (multi-stage vs. single-stage distillation).

Metric	With Layer Fusion	Without Layer Fusion	Progress	Possible Reason
Loss	0.6329	0.6873	−5.44%	Fused layers reduce redundancy and improve feature integration.
Accuracy	0.9606	0.9223	+3.83%	Combined hierarchical features capture richer representations.
Precision	0.9622	0.9303	+3.19%	Reduced noise in predictions due to fused feature maps.
Recall	0.9577	0.9159	+4.18%	Enhanced feature aggregation improves detection of true positives.
F1-score	0.9591	0.9233	+3.58%	Harmonized precision and recall from fused multi-scale information.
AUC	0.9905	0.9879	+0.26%	Stronger separability from fused layer's discriminative power.

Table 14. Effect of layer fusion (multi-stage distillation + fused layer).

Optimization Method	Model Size (KB)	Inference Latency (s / step)	Memory Usage (MB)
Hybrid Student Baseline	229,492	3	0.24
Knowledge Distillation (KD)	37,585	2	0.22
Reduction	191,907	1	0.02
Percentage	**83.62%**	**33.33%**	**8.33%**
Post-Training Quantization (PTQ)	18,778	41m	0.20
Reduction	210,714	2.95	0.04
Percentage	**91.81%**	**98.63%**	**16.67%**

Table 15. Comparison of quantization and knowledge distillation methods with size, latency, and memory efficiency.

respectively, due to better feature extraction and sensitivity to true positives. Furthermore, the AUC score for the multi-stage model (98.79%) slightly surpasses that of the single-stage model (98.57%), demonstrating improved class separability through multi-scale feature refinement.

2. **Effect of layer fusion**: Table 14 presents the effect of layer fusion in multi-stage knowledge distillation, demonstrating improvements in performance metrics. The inclusion of layer fusion reduces loss by 5.44% and enhances accuracy, precision, recall, and F1-score, with the highest improvement observed in recall (+4.18%) due to enhanced feature aggregation. Additionally, the AUC score increases to 99.05%, indicating stronger class separability and improved discrimination power from fused hierarchical features.
3. **Importance of post-training quantization**: The study compares two optimized methods to evaluate the effectiveness of reducing the size and parameters of the model to deploy on resource-constrained devices. Table 15 provides a comparative study of two prominent model optimization processes, Knowledge Distillation (KD) and Post-Training Quantization (PTQ), against the original Hybrid Student Baseline (HSB), concentrating on model size, inference latency, and memory efficiency. The unoptimized model inhabits 229,492 KB, demanding approximately 3 seconds per inference step and consuming 0.24 MB of memory. Applying Knowledge Distillation drastically reduced the size to 37,585 KB, achieving a size reduction of 83.62% and also a 33.33% reduction in inference latency (from 3s to 2s), with a fair 8.33% gain in memory usage, dropping from 0.24 MB to 0.22 MB. PTQ further stretched the boundaries of optimization. The PTQ compressed to the impression of 18,778 KB, a remarkable 91.81% reduction in size relative to the baseline. Additionally, it generated a 98.63% latency improvement, reducing the step time from 3 seconds to merely 41 milliseconds. Memory usage also decreased to 0.20 MB, a 6.67% decline. Figure 12 represents the trade-off between model size and AUC performance. Collectively, these findings demonstrate that both KD and PTQ substantially improve the model's deployability, with PTQ showing exceptional performance in latency-critical applications and KD proposing a balanced and efficient trade-off between AUC and efficiency.

Explainable AI analysis

This section deals with various XAI techniques to enhance model interpretability and provide valuable insights into the decision-making process. Fig. 13 depicts an input image and its corresponding visualizations generated

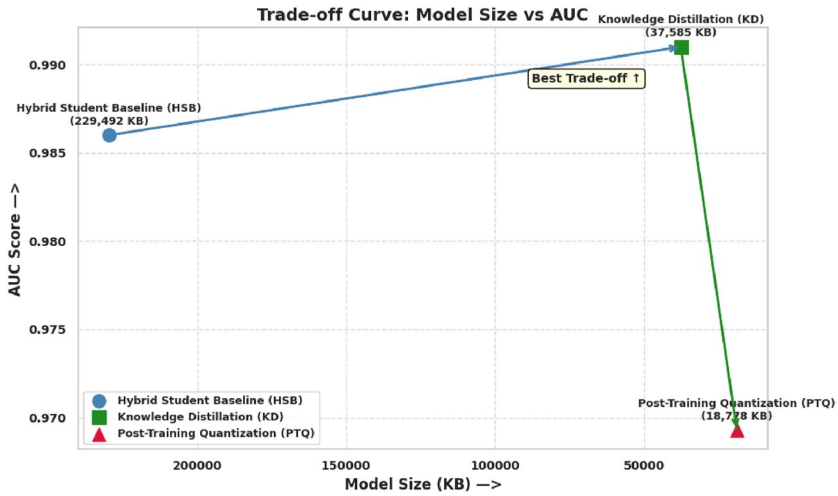


Fig. 12. Trade-off analysis between model compression and AUC performance.

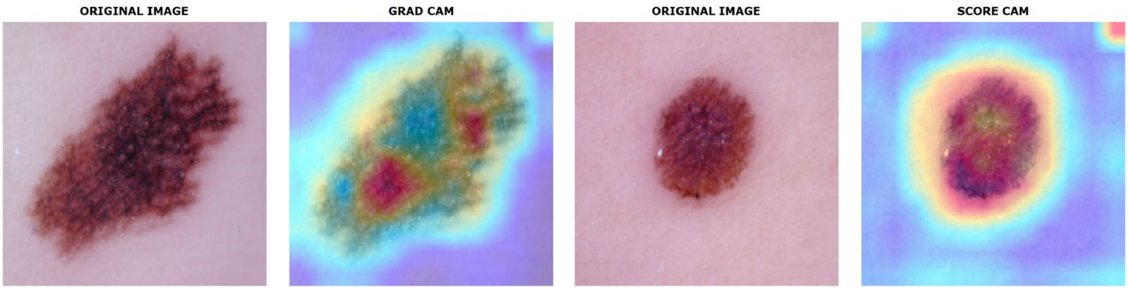


Fig. 13. Visualization of Grad-CAM and Score-CAM.

by Grad-CAM and Score-CAM. Grad-CAM highlights the regions with the gradient information that influenced the prediction by producing a heatmap. However, Score-CAM produces the heatmap by masking part of the image and measuring the resulting change in the model's confidence score without relying on gradients. Both visualizations aim to interpret the model decisions with overlaying activation maps of the sample image. Fig. 14 illustrates visualizations of LIME for six different images corresponding to the predicted class (0 to 7). LIME produces the corresponding interpretations by perturbing regions from the image and observing how this segmentation impacts the model performance in case of predictions. The highlighted areas indicate which regions influenced strongly in classification.

Limitations of the study

While the proposed multi-stage knowledge distillation technique achieved significant performance in skin cancer classification, several limitations must be acknowledged. These limitations include.

1. **Training Complexity and Optimization Challenges:** Multi-layer control increases training complexity, may cause incompatible gradients, and raises additional hyperparameters for tuning. Additionally, the implementation benefits tend to plateau beyond a particular level of supervision.
2. **Architectural and Resource Constraints:** There is a risk of over-regularization, problems in aligning heterogeneous teacher-student architectures, and growing computational and memory needs due to multi-layer activation handling.
3. **Generalization and Explainability Limitations:** MSKD processes oftentimes lack generalizability across diverse domains, stay underexplored in 3D imaging contexts, and could nicely incorporate cutting-edge explainability approaches for enhanced interpretability. A limitation of this study is that the explainability results from Grad-CAM, Score-CAM, and LIME were not validated by domain experts, making the assessment purely qualitative.

Addressing these challenges through innovations in architecture design and explainability methods is essential to fully unlock the potential of Multi-Stage Knowledge Distillation for practical, real-world deployments and to guide future research directions.

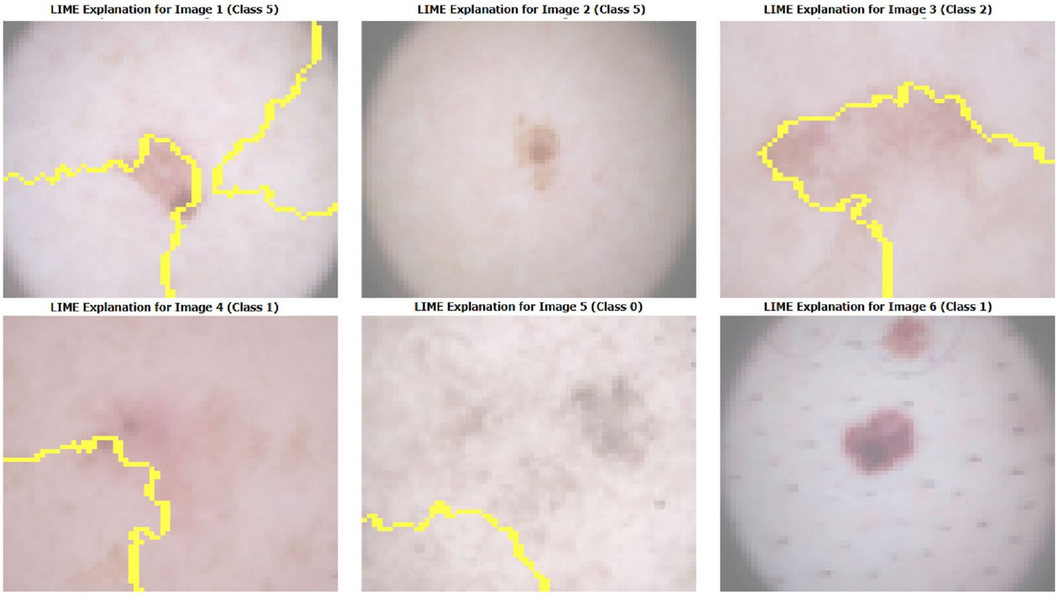


Fig. 14. Visualization of LIME.

Reference	Dataset	Models	Performance Metrics
18	ISIC 2017 + ISIC 2018 datasets	Hybrid AI framework with DenseNet121 and InceptionResNetV2	Accuracy: 93.04%, Precision: 93.0%, Recall: 92.0%, F1-score: 93.12%
19	Subject-specific datasets (2,541 images)	Modified AlexNet with VGG16 and VGG19	Accuracy improved from 94% to 97.18%
20	ISIC 2017	RAPS-based model selection with CNNs and Grey Wolf Optimizer	Best model achieved 94.5% accuracy
21	DermIS and ISIC 2018 datasets	PCA with AdaBoost and EfficientNet B0	Accuracy: 93% (DermIS), 91% (ISIC 2018)
22	ISIC 2018 dataset	Modified CNN with hierarchical features	Accuracy: 98.25%, Precision: 97.32%, Recall: 97.10%, F1-score: 96.92%
33	HAM10000	Hybrid system with Swin Transformer and DGSNLA Network	Accuracy: 94.21%, Precision: 96.62%, Recall: 96.25%, F1-score: 96.64%
34	ISIC 2018	ECRNet using CNN and ViT	Accuracy: 92.12%, Precision: 88.81% Recall: 86.05%, F1-score: 87.23%
26	ISIC 2019, HAM10000, Private dataset	EFAM-Net with ConvNeXt, PCNXt, and MEAFF	Accuracy: 92.30% (ISIC 2019), 93.95% (HAM10000), 94.31% (private dataset)
27	ISIC 2018, Mendeley dataset	DSCIMABNet with multi-head attention	Accuracy improved to 99.40% (from 84.28%) for ISIC 2018 and 99.37% (from 92.58%) for Mendeley
28	ISIC 2018, HAM10000	Max voting method using Random Forest, SVM, and MLP NN	Accuracy: 94.70%
44	ISIC dataset	Lightweight model using ConvNeXtV2 and focal self-attention	Accuracy: 93.60%, Precision: 91.69%, Recall: 90.05%, F1-score: 90.73%
45	ISIC 2018	Lightweight hybrid model using ConvNeXtV2 and focal self-attention	Accuracy: 93.48%, Precision: 93.24%, Recall: 90.70%, F1-score: 91.82%
25	HAM10000	Lightweight model for accurate skin cancer diagnosis	Accuracy: 98.54%, Precision: 98.59%, Recall: 98.61%, F1-score: 98.56%
46	Multimodal Derm7pt	MDSIS-Net	Accuracy: 96%, F1-score: 94.7%
47	Multimodal ISIC 2024	PMM-FL	Accuracy: 98.34%, F1-score: 97.65%
This Study	HAM10000	Multi-stage knowledge distillation with layer fusion via distilled models (CNN + EfficientNet)	Distilled model: Accuracy: 95.88%, Precision: 96.12%, Recall: 95.70%, F1-score: 95.91%

Table 16. Comparative analysis of related skin cancer classification with the proposed work.

Comparison with existing works

Table 16 presents a comparative analysis of various skin cancer classification models, highlighting their datasets, model architectures, and performance metrics. The proposed multi-stage knowledge distillation approach with layer fusion outperforms several prior studies, achieving an accuracy of 95.88%, precision of 96.12%, recall of 95.70%, and an F1-score of 95.91%. Compared to other models, the distilled network leveraging CNN and EfficientNet demonstrates superior classification performance, making it a promising approach for skin cancer detection.

Limitation Addressed	Mechanism by which MSKD Mitigates Limitation
Single-point supervision	Employs multiple intermediate feature alignments rather than leaning solely on final output logits.
Shallow knowledge transfer	Facilitate distillation of knowledge from various depths within the network order.
Neglect of structural information	Combine attention maps and embedding alignment to keep structural knowledge.
Teacher-student architectural mismatch	Utilizes adapter modules to moderate differences between heterogeneous network architectures.
Limited robustness and generalization	Layer-wise supervision improve the student's ability for potent feature learning and domain generalization.
Lack of interpretability	Intermediate supervision points improved visibility into the distillation process, assisting diagnostics and model transparency.

Table 17. Core limitations addressed by Multi-Stage Knowledge Distillation (MSKD) and its associated benefits.

Benefits of multi-stage knowledge distillation (MSKD)

Multi-Stage Knowledge Distillation (MSKD) has appeared as a promising strategy to overcome the constraints of traditional distillation methods by enabling more fine and flexible knowledge transfer via supervision at multiple intermediate layers. Table 17 summarizes critical limitations in traditional knowledge distillation and emphasizes how Multi-Stage Knowledge Distillation (MSKD) strategically mitigates them individually.

Conclusion

Skin cancer remains a critical global health concern, with diagnostic complexity arising from the morphological likenesses between malignant and benign lesions. Current outcomes in skin lesion analysis have shown significant promise in enhancing early diagnosis, prevention, and treatment effectiveness. Leveraging the HAM10000 dataset, a diverse representation of dermoscopic images, this study implements a robust pipeline, determining refined preprocessing and class-balancing augmentation to mitigate dataset biases. A hybrid teacher architecture containing Vision Transformer (ViT) and ConvNeXT backbones was employed to extract rich semantic features with increased discriminatory ability. The student network, composed of CNN and EfficientNet, was strategically selected to balance model accuracy with computational tractability, particularly for real-time and edge deployment contexts. A multi-stage knowledge distillation framework was introduced, allowing for the progressive transfer of hierarchical feature representations from the teacher to the student. This approach not only maintained diagnostic performance but also significantly diminished memory and computational requirements. Further enhancement was achieved through layer fusion techniques, which enabled synergistic integration of intermediate and deep-layer knowledge, yielding superior generalization on unseen samples. Post-training quantization was applied to compress the final student model, ensuring lightweight deployment without compromising diagnostic accuracy. Explainable AI (XAI) techniques, including Grad-CAM, Score-CAM, and LIME, are utilized to generate saliency-based heatmaps, providing critical transparency in model decision-making and increasing clinical trust.

Future directions

Future directions involve extending the framework to multimodal learning by integrating complementary datasets, adopting 3D dermoscopic image analysis, enhancing segmentation granularity, optimizing model pruning strategies, and employing advanced techniques to address class imbalance. The envisioned web-based application and edge deployment potential underscore the system's scalability and translational relevance. Future work will include expert-driven evaluation to quantitatively assess the clinical relevance of XAI-generated saliency maps. Runtime benchmarks on mobile or edge devices can be done to assess the real-world deployment feasibility of the proposed lightweight model.

Data availability

The datasets analyzed during the current study are publicly available. 1. HAM10000 (Human Against Machine with 10000 training images): This dataset is hosted on the Harvard Dataverse and includes dermatoscopic images of common pigmented skin lesions. It can be accessed at: https://doi.org/10.7910/DVN/DBW86T. 2. ISIC 2018 (International Skin Imaging Collaboration Challenge 2018): This dataset is part of a benchmark challenge for automated skin lesion analysis and includes dermoscopic images along with expert annotations for lesion segmentation and classification. It is publicly available at: https://challenge2018.isic-archive.com/.

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Author contributions
M.A.P. and R.A. wrote the main manuscript text and did validation, investigation, resources, and data curation, and G.K.O.M. prepared figures, resources, data curation, supervision, project administration, and funding acquisition, M.I, M.M and R.K. did investigation, resources, data curation. All authors reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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